

Linear Algebra (MT1004)

Sessional-I Exam

Date: September 21, 2026

Course Instructor(s)

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Rubric

Total Time (Hr): 1 Hour

Total Marks: 50

Total Questions: 1(5 parts)

Roll No

Section

Student Signature

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Instruction/Notes:

1. Programmable calculators are not allowed.
2. Marks will be awarded till the correct calculations, incorrect step voids further marking.
3. Attempt all question parts together. Question attempted in separate parts will not be marked.
4. Q#1(e) must be returned to the invigilator along with answer sheet.

CLO #1: Use concept of elementary row operations to find the inverse of square matrices, determinant of a matrix and solving the system of linear equations.

Q#1(a)[10]: Solve the following system of equations and determine the total number of possible solutions when $0 \leq \alpha \leq 2\pi$, $0 \leq \beta \leq 2\pi$, and $0 \leq \gamma \leq 2\pi$.

$$\begin{aligned}1 \sin \alpha + 2 \cos \beta + 3 \tan \gamma &= 0, \\2 \sin \alpha + 5 \cos \beta + 3 \tan \gamma &= 0, \\-1 \sin \alpha - 5 \cos \beta + 5 \tan \gamma &= 0.\end{aligned}$$

Hint: For convenience assume $x = \sin \alpha$, $y = \cos \beta$ and $z = \tan \gamma$ and then solve.

Solution:

By substituting $x = \sin \alpha$, $y = \cos \beta$, and $z = \tan \gamma$ so that the system becomes

$$\begin{aligned}x + 2y + 3z &= 0 \\2x + 5y + 3z &= 0 \\-x - 5y + 5z &= 0\end{aligned}$$

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 5 & 3 & 0 \\ -1 & -5 & 5 & 0 \end{bmatrix} \leftarrow \text{The augmented matrix for the system}$$

1 Mark

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$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & -3 & 8 & 0 \end{bmatrix} \leftarrow \begin{array}{l} -2 \text{ times the first row was added to the second row} \\ \text{and the first row was added to the third row.} \end{array}$$

3 Marks

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix} \leftarrow 3 \text{ times the second row was added to the third row.}$$

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \leftarrow \text{The third row was multiplied by } -1$$

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \leftarrow \begin{array}{l} 3 \text{ times the third row was added to the second row and} \\ -3 \text{ times the third row was added to the first row.} \end{array}$$

3 Marks

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \leftarrow -2 \text{ times the second row was added to the first row.}$$

This system has exactly one solution $x = 0$, $y = 0$, $z = 0$.

On the interval $0 \leq \alpha \leq 2\pi$, the equation $\sin \alpha = 0$ has three solutions: $\alpha = 0$, $\alpha = \pi$, and $\alpha = 2\pi$.

On the interval $0 \leq \beta \leq 2\pi$, the equation $\cos \beta = 0$ has two solutions: $\beta = \frac{\pi}{2}$ and $\beta = \frac{3\pi}{2}$.

On the interval $0 \leq \gamma \leq 2\pi$, the equation $\tan \gamma = 0$ has three solutions: $\gamma = 0$, $\gamma = \pi$, and $\gamma = 2\pi$.

Overall, $3 \cdot 2 \cdot 3 = 18$ solutions (α, β, γ) can be obtained by combining the values of α, β and γ

3 Marks

b)[10]: Define an elementary matrix? Use the **Inversion Algorithm** to find the inverse of

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$$

and also **write down & verify** A and A^{-1} as a product of elementary matrices, where

$$A = E_1^{-1} E_2^{-1} E_3^{-1} \dots E_{k-1}^{-1} E_k^{-1}, \quad A^{-1} = E_k E_{k-1} \dots E_3 E_2 E_1.$$

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Solution:

Elementary Matix:

An elementary matrix is a matrix obtained from an identity matrix by performing a single elementary row operation.

1 Marks

By Inversion Algorithm

$$[A | I] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 7 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & 1 \end{array} \right] \leftarrow -3 \text{ times the first row was added to the second row}$$

The corresponding elementary matrix is

$$E_1 = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}, \quad E_1^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$$

2 Marks

$$\left[\begin{array}{cc|cc} 1 & 0 & 7 & -2 \\ 0 & 1 & -3 & 1 \end{array} \right] \leftarrow -2 \text{ times the second row was added to the first row}$$

$$E_2 = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}, \quad E_2^{-1} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$$

3 Marks

Since $E_2 E_1 = I \Rightarrow A^{-1} = E_2 E_1$ and $A = E_1^{-1} E_2^{-1}$

$$A^{-1} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$$

2 Marks

$$A = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}$$

2 Marks

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c)[10]: Solve by **Cramer's Method**, any other method for solution will not be rewarded.

$$-2x_1 + x_3 = -32,$$

$$-x_2 - x_3 = 14,$$

$$x_1 + x_2 - 4x_3 = 11.$$

Solution:

By Cramer's Rule

$$A = \begin{bmatrix} -2 & 0 & 1 \\ 0 & -1 & -1 \\ 1 & 1 & -4 \end{bmatrix}, \quad b = \begin{bmatrix} -32 \\ 14 \\ 11 \end{bmatrix}$$

$$\det(A) = -2(4 + 1) + 1(1) = -9$$

1 Mark

$$A_1 = \begin{bmatrix} -32 & 0 & 1 \\ 14 & -1 & -1 \\ 11 & 1 & -4 \end{bmatrix}$$

$$\det(A_1) = -135$$

$$x_1 = \frac{\det(A_1)}{\det(A)} = \frac{-135}{-9}$$

$$x_1 = 15$$

3 Marks

$$A_2 = \begin{bmatrix} -2 & -32 & 1 \\ 0 & 14 & -1 \\ 1 & 11 & -4 \end{bmatrix}$$

$$\det(A_2) = 108$$

$$x_2 = \frac{\det(A_2)}{\det(A)} = \frac{108}{-9}$$

$$x_2 = -12$$

3 Marks

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$$A_3 = \begin{bmatrix} -2 & 0 & -32 \\ 0 & -1 & 14 \\ 1 & 1 & 11 \end{bmatrix}$$

$$\det(A_3) = 18$$

$$x_3 = \frac{\det(A_3)}{\det(A)} = \frac{18}{-9}$$

$$x_3 = -2$$

$$\text{Solution set} = \{x_1, x_2, x_3\} = \{15, -12, -2\}$$

3 Marks

d)[10]: Define co-factor of a square matrix. If A is an invertible matrix of order $n \times n$, then prove that $A (C_{ij})^T = \det(A) I_n$, where C_{ij} is a co-factor matrix of A .

Co-factor:

The co-factor C_{ij} of an entry a_{ij} in a square matrix A is defined as $C_{ij} = (-1)^{i+j} M_{ij}$, where M_{ij} is the minor determinant formed by deleting the i th row and j th column.

1 Marks

Proof:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \dots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

$$C_{ij} = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ \vdots & \vdots & & \vdots \\ c_{i1} & c_{i2} & \dots & c_{in} \\ \vdots & \vdots & & \vdots \\ c_{n1} & c_{n2} & \dots & c_{nn} \end{bmatrix}$$

$$(C_{ij})^t = \begin{bmatrix} c_{11} & \dots & c_{j1} & \dots & c_{n1} \\ c_{12} & \dots & c_{j2} & \dots & c_{n2} \\ \vdots & & \vdots & & \vdots \\ c_{1n} & \dots & c_{jn} & \dots & c_{nn} \end{bmatrix}$$

3 Marks

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Now consider

$$A(C_{ij})^t = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \dots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} c_{11} & \dots & c_{j1} & \dots & c_{n1} \\ c_{12} & \dots & c_{j2} & \dots & c_{n2} \\ \vdots & & \vdots & & \vdots \\ c_{1n} & \dots & c_{jn} & \dots & c_{nn} \end{bmatrix}$$

The entry in the i th row and j th column of the product $A(C_{ij})^t$ is

$$a_{i1} c_{j1} + a_{i2} c_{j2} + \dots + a_{in} c_{jn}$$

If $i = j$ then the above expression is the cofactor expansion of $\det(A)$ along the i th row of A . Otherwise zero.

$$= \begin{bmatrix} \det(A) & 0 & \dots & 0 \\ 0 & \det(A) & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & \det(A) \end{bmatrix}$$

3 Marks

$$= \det(A) \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} \text{ which is scalar multiplication of } \det(A) \text{ with } I_n$$

$$A(C_{ij})^t = \det(A) I_n$$

3 Marks

e)[10]: Fill in the blanks. (Only 10 correct will be rewarded.). Return this to the invigilator.

- Let A be a non-invertible $n \times n$ matrix, then the homogeneous system $A\mathbf{x} = \mathbf{0}$ has infinitely many solution(s).
- If the homogeneous system $A\mathbf{x} = \mathbf{0}$ has infinitely many /non-trivial solution, then the reduced row echelon form of A is not I_n .
- If the reduced row echelon form of A is I_n then A can be expressed as a product of elementary matrices.

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4. If A is invertible then $Ax = b$ has exactly one / a unique solution for every $n \times 1$ column matrix b .
5. If the reduced row echelon form of A is not I_n , then A is singular / non-invertible.
6. If A is an invertible matrix then $Adj(A)$ is equal to multiplicative inverse of A with the scale factor of $\det(A)$.
7. If an augmented matrix contains a row of the form $[0 \ 0 \ 0 \mid c]$, with $c \neq 0$, then the corresponding system will be inconsistent.
8. A square matrix A is called skew-symmetric if $A^T = -A$.
9. The determinant of any diagonal or triangular matrix is equal to the product of the entries on its main diagonal.
10. A homogeneous linear system $Ax = 0$ with more unknowns than equations always has infinitely many solution(s).
11. The sum of the main diagonal entries of a square matrix A is called the trace of A .
12. The cofactor C_{ij} of an entry a_{ij} is related to its minor M_{ij} by the formula $C_{ij} = (-1)^{i+j} M_{ij}$.
13. For any $m \times n$ matrix A , the product matrix AA^T is always a symmetric / square matrix.
14. According to Cramer's Rule, if $\det(A) \neq 0$, then the j th unknown variable x_j in $Ax = b$ is given by $x_j = \frac{\det(A_j)}{\det(A)}$.
15. Multiplying the entries in a row/column by the co-factors of the same row/column entries in a square matrix and then taking the sum will be equal to the determinant of a square matrix.
16. Multiplying the entries in a row/column by the co-factors of some different row/column entries in a square matrix and then taking the sum will be equal to zero.

Good Luck